Ensemble Learning for Robust Image Classification under Distribution Shift

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xAI-Proj-M: Domain Generalization for Robust Machine Learning
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Abstract

Robustness under distribution shift remains a central challenge in computer vision. While ensemble methods are widely used to improve generalization, their effectiveness depends on both model diversity and aggregation strategy. In this work, we compare homogeneous and heterogeneous ensembles under synthetic and real-world distribution shifts, and evaluate whether adaptive expert selection with a mixture-of-experts routing mechanism can better exploit model diversity

We construct a homogeneous CNN ensemble and a heterogeneous ensemble combining convolutional networks with a Vision Transformer. Performance is evaluated on clean data, synthetic corruption benchmarks, and a student-collected out-of-distribution dataset. Oracle analysis is used to quantify the potential benefit of complementary expertise.

Results show that homogeneous ensembles provide limited robustness gains due to correlated failures under shift. In contrast, heterogeneous ensembles exhibit larger oracle gaps and improved robustness in severe corruption regimes. However, a learned mixture-of-experts router primarily collapses to a dominant expert, yielding only modest improvements over uniform averaging. These findings indicate that ensemble robustness is constrained both by expert diversity and by the ability of routing mechanisms to effectively leverage that diversity. Our publicly accessible repository is available on GitHub.[1]

1 Introduction

Deep learning models for image classification are usually developed under the assumption that training and test data follow the same distribution, yet real deployments routinely violate this

[1] `https://github.com/deadPixelsGreta/xAI-proj-m-ws2526.` (accessed February 12, 2026).

condition. Changes in acquisition device, environment, or image quality induce distribution shifts that degrade performance, even for models that achieve high in-distribution accuracy (Quiñonero-Candela et al., 2008). This gap between benchmark performance and real-world reliability is a core challenge in domain generalization and out-of-distribution (OOD) robustness.

Improving clean accuracy alone does not guarantee robustness under domain shift, and modern vision models can fail abruptly on corrupted or unseen inputs while remaining highly confident in their predictions (Zhou et al., 2023). Such overconfident failures limit the applicability of deep models in safety- or reliability-critical applications and motivate methods that explicitly target robustness and calibration rather than only in-distribution accuracy.

Ensemble learning offers a practical way to improve robustness by combining models that fail in different ways under distribution shift (Dietterich, 2000). Heterogeneous ensembles that mix architectures tend to respond differently to the same domain shift than ensembles consisting of multiple instances of the same backbone (Petrakova et al., 2015). By aggregating predictions from such diverse experts, ensembles can achieve better robustness than single models or homogeneous ensembles (Wu et al., 2023).

Beyond static ensembling, mixture-of-experts (MoE) models introduce a routing mechanism that assigns inputs to experts in an input-dependent way (Masoudnia and Ebrahimpour, 2014). Rather than averaging all experts uniformly, a learned router can upweight experts that are better suited to a given input, potentially exploiting architectural diversity more efficiently under distribution shift. Yet it remains unclear whether routers trained on realistic data naturally specialize experts or instead collapse to using a single dominant model, leaving potential robustness gains untapped (Fedus et al., 2021).

In this work, we empirically study ensemble robustness under distribution shift with a focus on architectural diversity and expert routing. Concretely, we compare homogeneous ensembles of ResNet models with heterogeneous ensembles that combine multiple CNN backbones and a ViT, and we evaluate these ensembles on a subset of the ImageNet dataset under synthetic corruptions and on a custom phone-captured test set to quantify robustness under realistic distribution shifts. Our main contributions are: (1) a controlled comparison of homogeneous and heterogeneous ensembles under synthetic and real-world distribution shifts; (2) evidence that architectural heterogeneity improves robustness under domain shift, even when gains in clean accuracy are limited; and (3) an analysis of MoE routing behavior showing that, without explicit incentives for specialization, routing tends to over-rely on a single expert despite the presence of complementary models.

2 Background

2.1 Distribution Shift

In real-world application scenarios, training and testing conditions often differ in relevant properties. Such discrepancies are described within the framework of the dataset shift problem, which addresses the transfer of information between closely related but non-identical environments (Quiñonero-Candela et al., 2008, pp. 4–5). How reliable are predictions when evaluated on modified test data that do not originate from the training environment on which the model was trained?

Classical supervised learning methods are generally not designed to handle such distribution shifts. Predictive performance is optimized under the implicit assumption of stable training and testing conditions and can therefore exhibit substantial degradation even under moderate deviations in data distribution (Quiñonero-Candela et al., 2008, p. 28), Zhou et al. (2023). This observation highlights that high accuracy under training conditions alone is not a reliable indicator of robustness in real-world deployment scenarios and motivates approaches that explicitly target generalization under distribution shift.

2.2 Domain Generalization *(DG)* and Robustness *(RB)*

Many trained models achieve high performance on their respective training datasets but generalize insufficiently to data collected in different acquisition environments. (Torralba and Efros, 2011) demonstrate in an empirical analysis that commonly used image datasets contain strong dataset-specific biases, leading to significant performance drops in cross-dataset evaluations. Models therefore learn not only task-relevant features but also domain-specific regularities that are not stable outside the training domain.

Domain Generalization addresses this issue by explicitly considering learning under distributional differences across domains. In contrast to domain adaptation, it assumes that no data from the target domain are available during training (Muandet et al., 2013). The objective is to learn representations that remain stable across multiple source domains and can transfer to previously unseen domains.

Domain Generalization assumes that input distributions may differ across domains, while the underlying relationship between inputs and target labels remains stable. In other words, the marginal distribution of the inputs can change across domains, but the conditional relationship between input and label is assumed to be invariant. This means that although data may vary in appearance or acquisition conditions, the fundamental decision rule that maps inputs to outputs is shared across domains Muandet et al. (2013).

(Muandet et al., 2013) formulate domain generalization as the problem of extracting features from multiple related domains that are invariant to domain-specific variations while preserving the underlying functional relationship between input and target variable. DG therefore represents a central approach for achieving robustness to systematic distribution shifts without requiring post-hoc adaptation to new domains.

2.3 Ensemble Learning *(EL)*

Since DG and RB aim to prevent performance degradation under unknown distribution shifts, the question arises as to which methodological approaches are suitable for increasing robustness. In the survey by Zhou et al. (2023), ensemble learning is categorized as a model-based DG approach in which robustness is achieved by combining multiple models or model states. The underlying idea is that different models respond differently to distribution shifts and therefore do not fail on the same inputs. By aggregating multiple hypotheses, complementary error patterns can be exploited and reliance on a single model can be reduced.

The motivation is that different models exhibit different error patterns and thus do not fail on identical inputs. By aggregating multiple models, such complementary errors can be leveraged and the variance of individual predictions can be reduced.

In the context of DG and RB, ensemble learning becomes particularly relevant. Under distribution shifts, models often respond differently to new or altered data, such that no single model performs optimally across all domains. Ensembles provide a mechanism to increase robustness at the model level by considering multiple hypotheses and combining their predictions.

Based on this theoretical background and the available resources, the ensemble method was selected as the appropriate project approach. In particular, given the previously trained 18-layer and 50-layer ResNets, it was natural to investigate a homogeneous ensemble and contrast it with a more heterogeneous ensemble approach along the dimension of architectural diversity.

3 Methodology

3.1 Setup

The experiments were conducted using multiple computational setups. Training was performed on several Windows-based workstations equipped with NVIDIA A4000 GPUs, using appropriately configured development environments managed with `conda`. In addition, training runs were executed on Google Colab Pro, utilizing NVIDIA A100 and T4 GPUs. To ensure availability and persistence of data, results, and trained models, a pipeline integrating GitHub and Google Drive was established. For each training run, the two best-performing models as well as the final trained model were stored on Google Drive. Experiments were organized using Git branches, and the corresponding notebooks were retrieved from GitHub, modified when necessary, and versioned accordingly.

All training runs and evaluations were logged using the online experiment tracking service *Weights & Biases (WB)*[2]. The platform enabled rapid inspection of intermediate results and provided aggregated metrics to support further optimization.

We implemented several training setups. Initially, the goal was to train two models from the residual nets, namely 18-layer and 50-layer ResNet. These models served as an initial baseline architectures for subsequent development and comparison. During training, hyperparameter sweeps were conducted to identify the optimal learning rate and the corresponding batch size. The primary focus was on achieving a high *validation accuracy*. The resulting models were retained and reused in later stages of the project.

At this stage, the main objective was to become familiar with the training setup and to establish a functional workflow covering data transfer, model training with validation, and systematic evaluation of results.

3.2 Homogeneous Ensemble

Constructing an ensemble from multiple ResNet variants represents a methodologically sound approach. ResNets of different depths, such as the 18-layer, 34-layer, and 50-layer ResNet, differ in model capacity, representational depth, and optimization behavior. Although all models are based on the same architectural principle, He et al. (2016) demonstrate that meaningful diversity exists within the ResNet family.

This diversity arises from the differing number of residual blocks. These blocks enable hierarchical refinement of feature representations. Deeper models, particularly the 50-layer ResNet, can therefore learn more complex and abstract representations. Shallower models such as the 18-layer ResNet, in contrast, learn different activation patterns and are more strongly characterized by localized transformations. As a result, the models exhibit different inductive biases and weight different aspects of the same input differently.

He et al. (2016) further show that ResNets of varying depth not only achieve different accuracy levels but also exhibit distinct error characteristics. These errors do not occur on the same inputs but are partially complementary. This is critical for ensemble usage, as incorrect predictions from individual models can be compensated by correct predictions from others. The strong performance of individual models can therefore be leveraged more effectively within an ensemble than through the deployment of a single deep model.

Since all models belong to the same architectural class and are trained under identical conditions on the same dataset, the ensemble remains homogeneous and interpretable. A ResNet ensemble is therefore particularly suitable as a baseline for systematically investigating ensemble effects, model diversity, and robustness.

Moreover, a ResNet ensemble can be directly compared to a single 50-layer ResNet. Because the 50-layer ResNet represents the strongest individual model within the ensemble, this comparison enables a clear attribution of ensemble effects. Performance gains achieved by the ensemble can thus be directly attributed to model combination rather than merely increased model capacity. This allows for a clean separation between improvements due to ensembling and those resulting solely from the use of a strong individual model.

From the perspective of ensemble learning, combining multiple models with different inductive biases is particularly well-suited for reducing variance and improving robustness. Shallower ResNet variants tend to rely more heavily on local and low-level visual patterns, whereas deeper ResNets learn more abstract and semantically rich representations. An ensemble of ResNet models with varying depths therefore enables controlled prediction diversity without altering the underlying architectural class. This makes ResNet-based ensembles particularly suitable as a foundation for analyzing robustness and error diversity under distribution shift.

[2]Weights & Biases: Machine learning experiment tracking. https://wandb.ai/site (accessed February 10, 2026).

3.3 Heterogeneous Ensemble

To test the effect of architectural diversity, we construct a heterogeneous ensemble from four backbones with distinct biases, all fine-tuned on the same ImageNet-subset training data.

These backbones are summarized in Table 1. They were specifically chosen to maximize inductive bias diversity: the CNNs provide complementary local feature extraction (texture, details, scaling), while the Vision Transformer (ViT) adds global reasoning. Identical training ensures differences arise from architecture alone.

Table 1: Backbone architectures comprising the heterogeneous ensemble. Each model is initialized with ImageNet-pretrained weights and fine-tuned on the same training data to isolate architectural inductive bias effects.

Backbone	Key characteristics	Reference
ResNet-34	strong local texture bias, stable baseline	He et al. (2016)
DenseNet-121	Dense connectivity for feature reuse	Huang et al. (2017)
EfficientNet-B0	Parameter-efficient, balanced scaling	Tan and Le (2019)
ViT-B/16	Global self-attention, shape-focused	Dosovitskiy et al. (2020)

Predictions are aggregated via simple averaging:

$$\hat{y} = \frac{1}{K} \sum_{k=1}^{K} p_k(\mathbf{x})$$

where $p_k(\mathbf{x})$ is the softmax output of expert k, and $K = 4$. This uniform weighting provides a baseline before introducing learned routing in Section 3.4.

Prior work suggests that architectural diversity can induce complementary failure modes under distribution shift (Wu et al., 2023). We evaluate this hypothesis empirically.

3.4 Mixture-of-Experts Router

To adaptively exploit architectural diversity, we extend the heterogeneous ensemble with a mixture-of-experts (MoE) router (Masoudnia and Ebrahimpour, 2014). Rather than uniformly averaging predictions, the router produces input-dependent weights over experts.

The routing network is a lightweight two-layer MLP that takes penultimate-layer feature representations from the ViT-B/16 backbone as input and outputs a probability distribution over experts via softmax. ViT features are used as routing input due to their global, content-aware representations, which are well-suited for identifying image characteristics associated with different failure modes.

Given gating weights $\mathbf{g} \in \Delta^3$, the final prediction is computed as

$$\hat{y} = \sum_{k} g_k p_k(\mathbf{x}).$$

The router is trained using cross-entropy loss on an 80/20 split of validation set, while all expert backbones remain frozen. Strong regularization, including heavy dropout, is applied to reduce the risk of expert collapse, as observed in large-scale MoE systems (Fedus et al., 2021).

This setup allows us to directly study whether expert diversity alone is sufficient for effective routing, or whether additional incentives for specialization are required.

4 Experimental Setup

4.1 Datasets

Experiments are conducted on a 10-class object classification task provided within the xAI-Proj-M framework. Standard evaluation is applied on the clean set, while robustness is evaluated under two types of distribution shifts.

Clean Dataset. The clean training set consists of 12.500 images, split to 1.250 images per class. A separate clean validation set is used for model selection and router training, consisting of 500 images (50 per class). All models are trained exclusively on the clean training split.

Synthetic Corruptions. The synthetic corruption benchmark consists of 500 images per corruption type (Gaussian noise and pixelation), evaluated across five severity levels. The dataset covers the same 10 object classes as the clean validation set. These perturbations modify low-level image statistics while preserving semantic content.

Real-World OOD Dataset. We evaluate models on a student-collected dataset containing approximately 300 images per student (30 per class), captured in outdoor or atypical environments using different devices. This dataset introduces shifts in background, lighting, viewpoint, and acquisition conditions.

All splits are strictly separated. No target-domain adaptation or corruption data is used during training.

4.2 Training Protocol

All backbone models are initialized with ImageNet-pretrained weights and fine-tuned on the clean training set. Optimization uses stochastic gradient descent (SGD) with momentum 0.9 and weight decay 10^{-4}. Training runs for up to 30 epochs with early stopping based on best validation accuracy (patience = 5 epochs). Backbone-specific hyperparameter optimization is described in the respective subsections.

For the **homogeneous ensemble**, a two-stage optimization strategy is employed.

In Phase 1, each Resnet is trained independently. Learning rate is optimized via Bayesian optimization over a log-uniform range $[10^{-5}, 10^{-1}]$, while batch size is selected from predefined candidate values. Early termination is implemented using Hyperband. All runs use cosine learning rate scheduling, weight decay 10^{-4}, label smoothing (0.1), and a fixed random seed (42).

In Phase 2, the learning rate search range is narrowed around the best configuration identified in Phase 1, and batch size candidates are restricted to the two best-performing values. Bayesian optimization with Hyperband early stopping is again employed.

The final homogeneous ensemble consists of the best-performing 18-layer, 34-layer and 50-layer ResNet models obtained after Phase 2. For a given input x, each model produces a class probability distribution $p_m(y \mid x)$ via softmax. The ensemble prediction is computed through uniform probability averaging,

$$p_{\text{ens}}(y \mid x) = \frac{1}{M} \sum_{m=1}^{M} p_m(y \mid x),$$

where $M = 3$ denotes the number of ensemble members. The final prediction corresponds to $\arg\max_y p_{\text{ens}}(y \mid x)$. No weighting or routing mechanism is applied in this setup in order to isolate the effect of architectural depth and augmentation-induced diversity.

For the **heterogeneous ensemble** a single, identical data augmentation pipeline is applied to all backbones to ensure fair comparison. Augmentations include RandomResizedCrop, horizontal flipping, and ColorJitter. Keeping the augmentation strategy fixed helps isolate architectural inductive bias as the primary source of diversity in the heterogeneous ensemble.

For the mixture-of-experts setup, backbone weights are frozen. The routing network is trained using cross-entropy on an 80/20 split of validation data to prevent evaluation leakage. No corruption or phone-captured data is used during training.

4.3 Evaluation Metrics

We evaluate all models using classification accuracy, calibration metrics, and oracle analysis. While standard accuracy measures realized performance, oracle accuracy quantifies the theoretical upper bound achievable by the ensemble.

Accuracy. We report overall classification accuracy on clean validation data, synthetic corruption benchmarks, and the phone-captured OOD dataset. For corruption experiments, performance is analyzed across severity levels to assess degradation under increasing shift.

Expected Calibration Error (ECE). Calibration quality is measured using ECE (Guo et al., 2017) computed on the ensemble's final prediction. For each sample, expert probabilities are first combined (via uniform averaging or routing weights), and the maximum predicted probability of the aggregated distribution is taken as the ensemble confidence. Predictions are grouped into 15 equally spaced confidence bins, and ECE is computed as the weighted average of the absolute difference between mean confidence and empirical accuracy within each bin. Lower ECE indicates better alignment between predicted probabilities and observed correctness.

Oracle Accuracy. To assess the potential benefit of expert diversity, we compute oracle accuracy. For each input sample, the oracle selects the expert that predicts the correct class if at least one expert is correct. Formally, oracle accuracy is defined as:

$$\text{Oracle} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}\Big(\exists k \text{ s.t. } \hat{y}_k(\mathbf{x}_i) = y_i\Big),$$

where $\hat{y}_k(\mathbf{x}_i)$ denotes the predicted class of expert k for sample $\mathbf{x}_i$.

Oracle accuracy therefore represents the maximum achievable performance under perfect expert selection. The gap between ensemble accuracy and oracle accuracy reflects the extent of complementary expertise within the ensemble. A small gap indicates correlated errors, whereas a large gap suggests substantial specialization potential.

In our analysis (see Section 5), oracle accuracy is used to quantify whether architectural heterogeneity creates exploitable diversity and whether the mixture-of-experts router successfully approaches this theoretical upper bound.

5 Results

5.1 Clean Validation Performance

As shown in Table 2, on the clean validation set, the **homogeneous ensemble** achieves an overall accuracy of 89.20%, compared to 88.20% (18-layer ResNet), 86.00% (34-layer ResNet), and 89.40% (50-layer ResNet). Thus, the ensemble performs comparably to the strongest individual backbone but does not surpass 50-layer ResNet in overall accuracy. Per-class analysis reveals that the ensemble matches or improves upon individual models in several categories (e.g., soup_bowl at 100.0%) while remaining close to the best single-model performance in others. Oracle accuracy reaches 93.60%, indicating additional potential gains if expert selection were optimal. The gap between ensemble and oracle accuracy amounts to 4.4 percentage points. The Expected Calibration Error (ECE) is 0.1458. The average disagreement between ensemble members is 0.0187, corresponding to an agreement ratio of 93.6%.

The **heterogeneous ensemble** achieves strong accuracy (see Table 3) on the clean dataset. Oracle accuracy reaches 96.20%, exceeding the realized ensemble performance. The moderate oracle gap indicates the presence of complementary expertise among backbones. However, uniform averaging already captures most of the attainable performance. Calibration analysis shows that ensemble-level ECE remains non-negligible, indicating that improved accuracy does not automatically imply improved probability calibration.

5.2 Corruption Set

Performance under synthetic corruption is evaluated across five severity levels for Gaussian noise and pixelation.

Figures 1 and 2 show accuracy as corruption severity increases for Gaussian noise and pixelation.

At low corruption levels, both ensembles and the strongest single models perform similarly. Differences are small and remain within a narrow range.

As severity increases, performance begins to diverge. Under Gaussian noise, the homogeneous ResNet ensemble follows the degradation pattern of the single ResNet-50 and exhibits a pronounced drop at higher severity levels. In contrast, the heterogeneous ensemble shows a flatter decline and maintains higher accuracy under severe corruption. The difference becomes most visible at severity levels 4 and 5.

A similar pattern is observed for pixelation. At low levels, both ensembles behave comparably. With increasing distortion, however, the homogeneous ensemble degrades more strongly, while the heterogeneous ensemble remains more stable.

Across both corruption types, the main signal lies in the slope of degradation rather than in absolute accuracy at low severity. Architectural diversity reduces sensitivity to increasing corruption strength, whereas homogeneous ResNet models tend to fail in a more correlated manner under severe shift.

5.3 Data Phone Set / Domain Shift

On the OOD dataset, shown in Table 4, the **homogeneous ensemble** achieves an overall accuracy of 55.8%, compared to 51.1% (18-layer ResNet), 51.6% (34-layer ResNet), and 56.3% (50-layer ResNet). As on the clean validation set, the ensemble performs comparably to the strongest single backbone but does not exceed 50-layer ResNet in overall accuracy. Per-class performance varies substantially across categories, with the ensemble achieving its highest accuracy on *wooden_spoon* (81.2%) and lowest on *soup_bowl* (34.3%). Oracle accuracy reaches 66.75%, resulting in a gap of 10.95 percentage points between ensemble and oracle performance. The Expected Calibration Error (ECE) is 0.0502. The average disagreement between ensemble members increases to 0.0289, corresponding to an agreement ratio of 78.7%.

Performance of the **heterogeneous ensemble** on the phone-captured dataset (see Table 5) exhibits a substantial drop relative to clean validation results, reflecting the severity of real-world distribution shift. Introducing architectural diversity improves realized ensemble accuracy compared to homogeneous CNN aggregation. More importantly, oracle accuracy increases to 72.49%, substantially exceeding realized ensemble performance. The magnitude of this oracle gap is notably larger than on clean data. This indicates that under real-world shift, different experts succeed on different subsets of the data. In other words, complementary expertise becomes more pronounced as the distribution diverges from training conditions. Despite this increased diversity, uniform averaging fails to recover oracle performance. The ensemble remains significantly below its theoretical upper bound. This suggests that the heterogeneous ensemble is not capacity-limited, but limited by its aggregation strategy at inference time.

The presence of substantial unrealized oracle potential motivates the introduction of an adaptive routing mechanism, which aims to exploit input-dependent specialization among experts.

5.4 Mixture-of-Experts Routing

We evaluate whether adaptive expert weighting can reduce the oracle gap observed on the phone-captured dataset. Figure 3 compares three configurations: (i) uniform averaging of the heterogeneous ensemble, (ii) the strongest single backbone (ViT-B/16), and (iii) the learned mixture-of-experts (MoE) router.

Uniform averaging treats all experts equally, regardless of input characteristics. Under real-world shift, this strategy underperforms the strongest single model, indicating that naive aggregation does not fully exploit architectural diversity. Replacing uniform averaging with a learned routing mechanism improves accuracy by approximately 2.4 percentage points relative to simple averaging. This gain is achieved without increasing model capacity or retraining the backbone networks. The improvement therefore reflects more effective utilization of existing expert diversity.

Despite this improvement, routed performance remains substantially below oracle accuracy. A significant portion of the ensemble's theoretical upper bound remains unrealized.

Routing Behavior Analysis. To understand how the router achieves its improvement, we analyze the distribution of learned gating weights. Rather than producing a balanced mixture of experts, the router assigns dominant probability mass to a single expert (see Figure 4). Approximately 85% of the routing weight is allocated to the ViT backbone, with the remaining CNN experts contributing minimally. This behavior indicates that the routing mechanism does not learn fine-grained, input-dependent specialization across experts. As shown in Figure 5, 60.1% of samples are classified correctly by the chosen expert. Missed oracle cases account for 12.4% of samples, representing the only region where improved routing could increase performance without modifying the ensemble. In contrast, 27.5% of samples correspond to hard failures, where no expert predicts correctly.

Class-level Routing Errors. Furthermore, we examine missed oracle cases in which the router selects the ViT backbone despite another expert predicting correctly. Figure 6 shows that these missed opportunities are not uniformly distributed across classes. They are concentrated in specific categories, including *mouse*, *notebook*, and *teapot*. This class-dependent pattern indicates that routing errors are structured rather than randomly distributed across the label space.

ViT Failure Analysis. To analyze routing blind spots, we compare image characteristics between samples where the ViT backbone predicts correctly and samples where it fails. Figure 7 shows that ViT failure cases exhibit systematically higher blur levels. On average, failure images are approximately 32% blurrier than successful predictions. Failure cases also exhibit lower contrast. Importantly, many of these blurry samples are correctly classified by CNN-based experts. For example, DenseNet-121 and EfficientNet-B0 jointly recover a substantial fraction of ViT failures on the *mouse* class. These observations indicate that routing errors are not caused by a lack of alternative experts. Instead, viable CNN predictions exist in regimes where the ViT backbone degrades under reduced image quality.

6 Discussion

6.1 Central Observations

Our findings highlight three central observations regarding ensemble robustness under distribution shift.

First, ensemble improvements depend on diversity among individual model errors (Dietterich, 2000). Uniform averaging provides limited gains when model errors are highly correlated. The oracle analysis demonstrates that robustness benefits depend on the presence of disagreement under shift. When experts succeed on different subsets of samples, ensemble potential increases; when they fail together, aggregation offers little advantage.

Second, architectural diversity is a primary driver of such complementary failure modes. Under real-world shift, heterogeneous ensembles exhibit larger oracle gaps than homogeneous CNN ensembles. This suggests that distinct inductive biases degrade differently under distribution changes. As a result, heterogeneous ensembles maintain robustness in regimes where homogeneous ensembles experience correlated collapse.

Third, our mixture-of-experts experiments reveal a limitation of naive routing strategies. Although routing improves performance relative to uniform averaging, the learned gating mechanism concentrates the majority of its weight on a single dominant expert. The breakdown of routing decisions shows that only a limited fraction of errors correspond to oracle cases, while a substantial portion arises from hard failures where no expert is correct.

Taken together, these results suggest that architectural diversity alone is insufficient to guarantee effective specialization. In the absence of explicit incentives, routing mechanisms tend to prioritize globally strong experts rather than conditionally coordinating complementary ones. Consequently, ensemble robustness under domain shift is constrained both by the diversity of available experts and by the ability of the aggregating mechanisms to exploit that diversity.

Future work may investigate routing objectives that explicitly encourage expert specialization or incorporate input-level uncertainty cues. Alternatively, increasing expert diversity through data-centric or representation-level strategies may reduce the proportion of hard failures observed under real-world shift.

6.2 Limitations

Several limitations contextualize our findings.

First, backbone representations are not adapted to target domains. All experts are trained exclusively on the clean in-domain dataset. Consequently, observed robustness gains arise purely from aggregation strategies rather than improved feature learning. It remains unclear whether joint adaptation or domain-aware representation learning would reduce the proportion of hard failures observed under real-world shift.

Second, the routing network is trained on a relatively small validation split. The limited training data may constrain the router's ability to learn fine-grained specialization and likely contributes to the observed concentration of routing weights on a single dominant expert. A larger or more diverse routing dataset may enable more conditional coordination across experts.

Third, the synthetic corruption benchmark (Gaussian noise and pixelation) captures low-level distribution shifts but does not model semantic or contextual variation. Real-world domain shift often involves changes in background, object context, and acquisition conditions that extend beyond pixel-level distortions.

Taken together, these limitations indicate that both representation quality and expert diversity jointly constrain ensemble robustness under domain shift.

6.3 Conclusion

This work examined ensemble robustness under distribution shift and provides three central insights. First, robustness gains are not a function of model count but of failure diversity. Ensembles improve only when experts make complementary errors rather than correlated ones.

Second, architectural heterogeneity primarily benefits robustness under shift, not clean accuracy. Distinct inductive biases degrade differently, creating complementary behavior that can be exploited under corruption and real-world variation.

Third, routing mechanisms do not automatically induce specialization. In our setting, the mixture-of-experts model behaves largely as expert selection rather than coordinated collaboration. Without explicit incentives, gating mechanisms favor globally strong experts instead of conditionally combining complementary ones.

Together, these findings suggest that ensemble robustness emerges less from stronger individual models and more from structured diversity and effective aggregation. Designing ensembles for robustness therefore requires intentional diversity engineering and routing objectives that promote specialization rather than dominance.

Author Contributions

Sebastian implemented and evaluated the homogeneous ensemble baseline, including training, validation, and corruption robustness analysis. Sebastian wrote the Background section, the homogeneous ensemble subsection of Methods, and contributed to the Experimental Setup and Results sections. Sebastian also authored the Conclusion.

Stephanus was responsible for implementation of the heterogeneous ensemble and the mixture-of-experts routing mechanism. This included backbone selection, routing architecture design, experimental execution, and routing-behavior analysis. Stephanus wrote the Introduction, the heterogeneous ensemble and routing subsections of Methods, the corresponding Results subsections, and the Discussion and Limitations sections.

Both authors jointly designed the experimental setup, reviewed all sections of the report, and contributed to interpretation of results and manuscript revisions.

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Declaration of Authorship

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Bamberg, February 12, 2026

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A Appendix

Table 2: Per-class accuracy (%) on the clean dataset. Columns compare the homogeneous ensemble with its constituent backbones (ResNet-18, ResNet-34, and ResNet-50). The overall row reports aggregate accuracy across classes.

Class	Ensemble	ResNet-18	ResNet-34	ResNet-50
binder	**98.0**	96.0	86.0	**98.0**
coffee_mug	78.0	78.0	76.0	**82.0**
computer_keyboard	**82.0**	**82.0**	78.0	76.0
mouse	78.0	66.0	**82.0**	78.0
notebook	78.0	**86.0**	80.0	82.0
remote_control	94.0	90.0	92.0	**96.0**
soup_bowl	**100.0**	98.0	96.0	98.0
teapot	98.0	**100.0**	96.0	98.0
toilet_tissue	90.0	**92.0**	86.0	88.0
wooden_spoon	96.0	94.0	88.0	**98.0**
Overall	89.2	88.2	86.0	**89.4**

Table 3: Per-class accuracy (%) on the clean dataset. Columns compare the heterogeneous ensemble with its constituent backbones (DenseNet-121, EfficientNet-B0, ResNet-34, and ViT-B/16). The overall row reports aggregate accuracy across classes.

Class	Ensemble	DenseNet121	EfficientNet	ResNet-34	ViT-B/16
binder	**100.0**	94.0	96.0	94.0	**100.0**
coffee_mug	82.0	78.0	76.0	82.0	**84.0**
computer_keyboard	86.0	76.0	76.0	80.0	**88.0**
mouse	80.0	74.0	82.0	72.0	**86.0**
notebook	84.0	82.0	**86.0**	82.0	80.0
remote_control	**96.0**	94.0	**96.0**	88.0	**96.0**
soup_bowl	**98.0**	**98.0**	96.0	**98.0**	94.0
teapot	**100.0**	98.0	**100.0**	96.0	**100.0**
toilet_tissue	**96.0**	92.0	94.0	92.0	94.0
wooden_spoon	**96.0**	**96.0**	94.0	94.0	94.0
Overall	**91.8**	88.2	89.6	87.8	91.6

Table 4: Per-class accuracy (%) on the phone-captured out-of-distribution dataset. Columns compare the homogeneous ensemble with its constituent backbones (ResNet-18, ResNet-34, and ResNet-50). The overall row reports aggregate accuracy across classes.

Class	Ensemble	ResNet-18	ResNet-34	ResNet-50
binder	55.3	51.5	45.6	**58.4**
coffee_mug	48.5	42.7	39.7	**54.9**
computer_keyboard	42.6	39.6	36.8	**44.7**
mouse	52.9	46.7	**56.3**	48.7
notebook	**66.0**	62.5	60.0	64.9
remote_control	62.7	52.9	58.8	**64.0**
soup_bowl	34.3	30.4	30.8	**35.3**
teapot	37.1	32.7	**40.0**	35.8
toilet_tissue	77.9	**78.8**	70.3	75.1
wooden_spoon	81.2	74.1	78.2	**81.4**
Overall	55.8	51.1	51.6	**56.3**

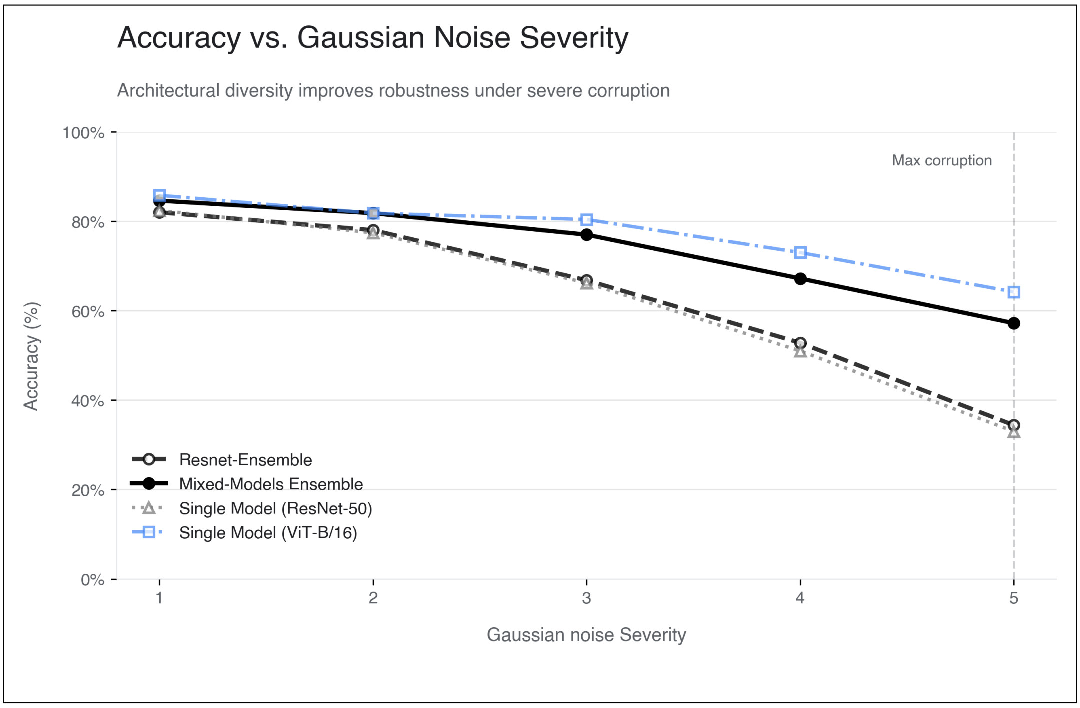

Figure 1: Classification accuracy under Gaussian noise corruption across five severity levels. Results are shown for a homogeneous ResNet ensemble, the heterogeneous mixed-model ensemble, and the strongest single backbones (ResNet-50 and ViT-B/16). Evaluation is performed on the synthetic corruption benchmark.

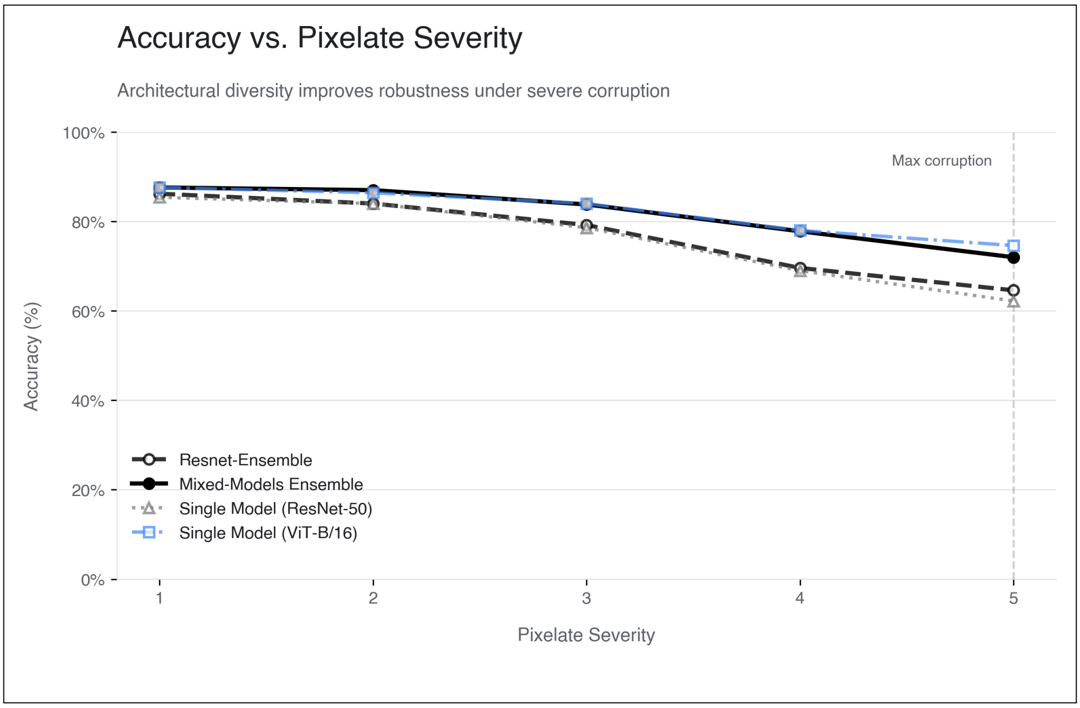

Figure 2: Classification accuracy under pixelation corruption across five severity levels. Results are shown for a homogeneous ResNet ensemble, the heterogeneous mixed-model ensemble, and the strongest single backbones (ResNet-50 and ViT-B/16). All evaluations are conducted on the synthetic corruption benchmark.

Table 5: Per-class accuracy (%) on the phone-captured out-of-distribution dataset. Columns compare the heterogeneous ensemble with its constituent backbones (DenseNet-121, EfficientNet-B0, ResNet-34, and ViT-B/16). The overall row reports aggregate accuracy across classes.

Class	Ensemble	DenseNet121	EfficientNet	ResNet-34	ViT-B/16
binder	56.9	38.2	51.7	48.5	**68.3**
coffee_mug	49.1	48.5	41.2	38.6	**57.1**
computer_keyboard	44.3	39.3	33.0	37.7	**54.8**
mouse	49.2	46.9	**49.4**	41.9	43.5
notebook	74.0	71.9	**74.5**	62.3	67.0
remote_control	57.5	54.4	55.7	44.3	**60.5**
soup_bowl	**32.5**	29.7	30.6	24.8	32.0
teapot	44.8	38.0	41.7	40.6	**49.2**
toilet_tissue	**86.4**	82.5	74.0	80.6	84.3
wooden_spoon	**86.0**	81.4	84.6	81.2	84.6
Overall	58.0	53.0	53.6	49.9	**60.1**

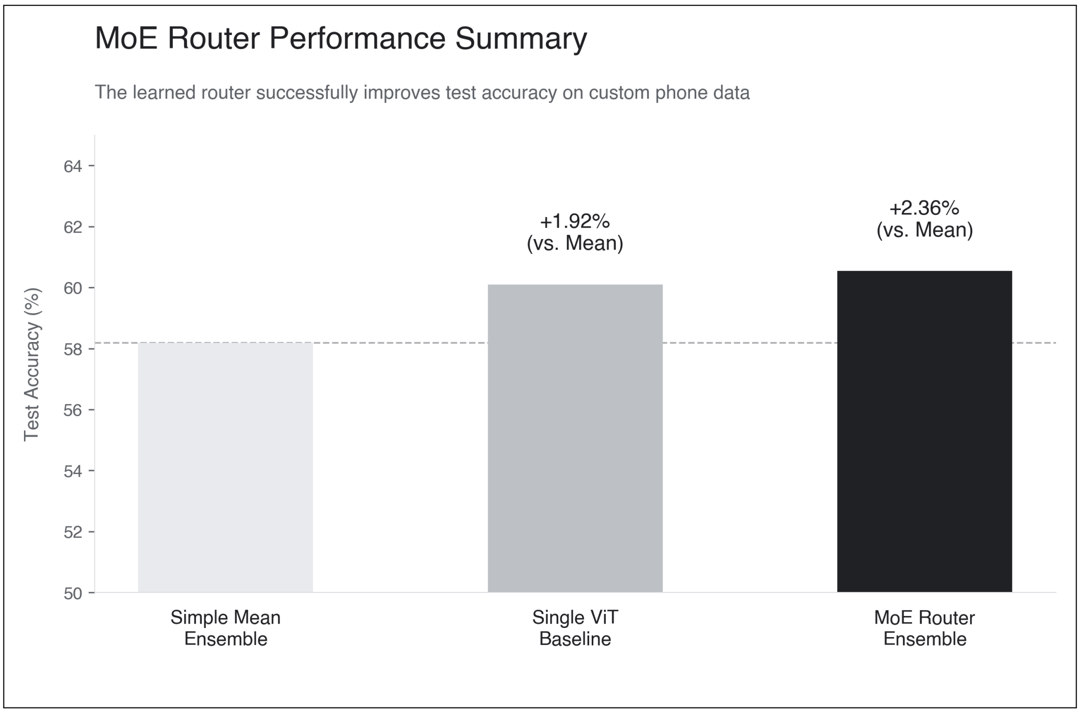

Figure 3: Test accuracy on the phone-captured dataset for uniform averaging (Simple Mean Ensemble), the strongest single backbone (ViT-B/16), and the Mixture-of-Experts (MoE) router. Percentage annotations indicate relative improvement over uniform averaging.

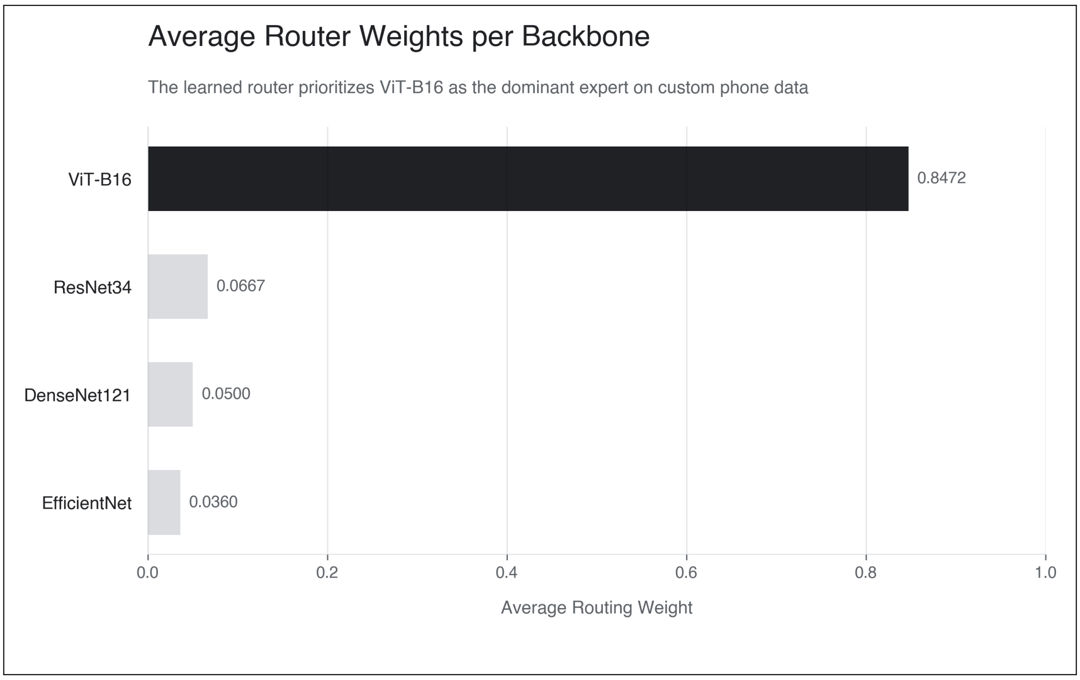

Figure 4: Average routing weight assigned to each backbone by the mixture-of-experts router on the phone-captured dataset. Weights are averaged across all samples. The router assigns the majority of probability mass to the ViT-B/16 backbone.

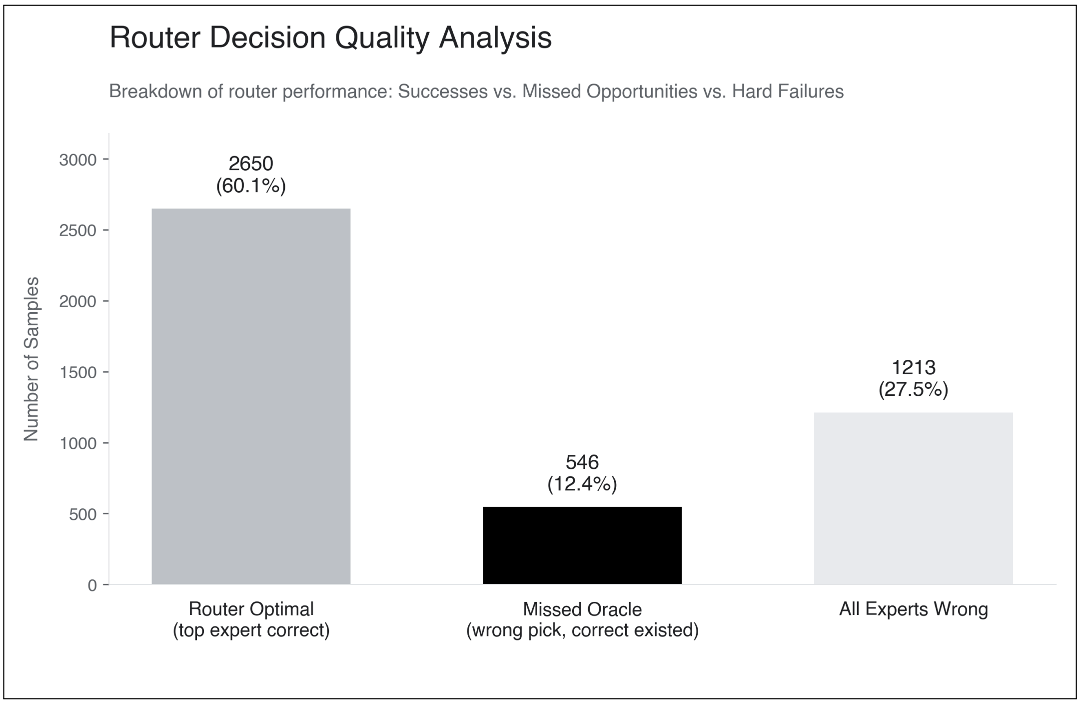

Figure 5: Breakdown of routing outcomes on the phone-captured dataset. Samples are categorized as: (i) router selects a correct expert (“Router Optimal”), (ii) router selects an incorrect expert although another expert predicts correctly (“Missed Oracle”), and (iii) all experts predict incorrectly (“All Experts Wrong”). Percentages indicate the proportion of total evaluation samples in each category.

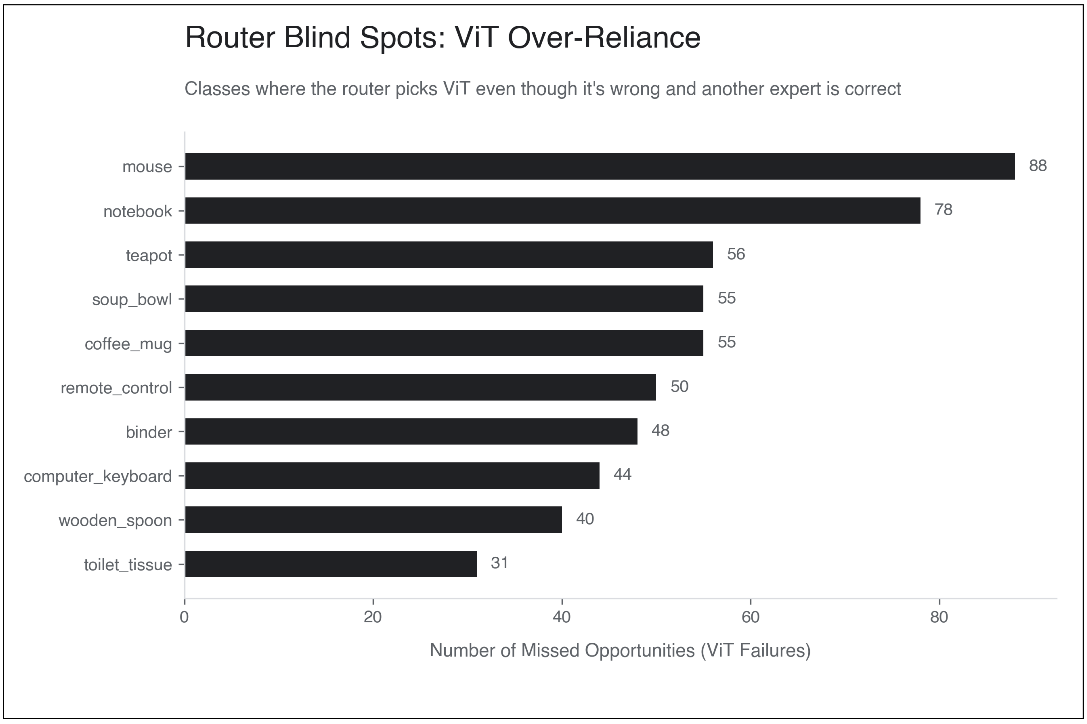

Figure 6: Class-wise distribution of missed oracle cases on the phone-captured dataset. The bars indicate the number of samples per class where the router selects the ViT backbone despite another expert predicting correctly.

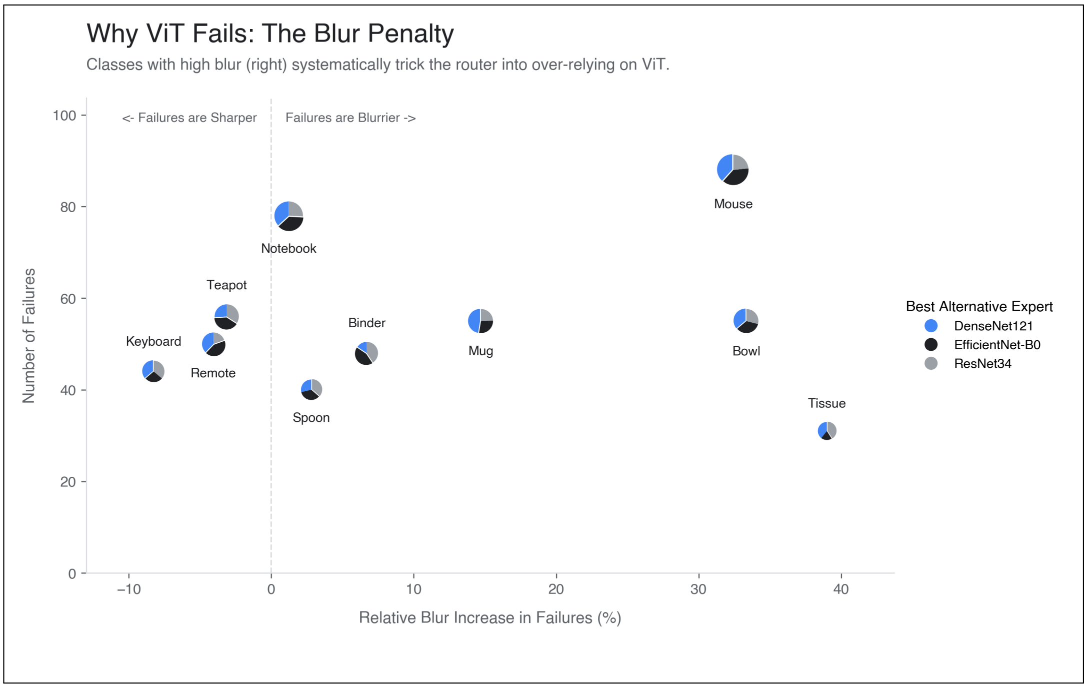

Figure 7: Analysis of ViT failure characteristics on the phone-captured dataset. The x-axis shows the relative increase in blur for failure cases compared to successful predictions, and the y-axis indicates the number of failures per class. Pie charts denote the alternative expert (DenseNet-121, EfficientNet-B0, or ResNet-34) that correctly predicts the label in these cases.